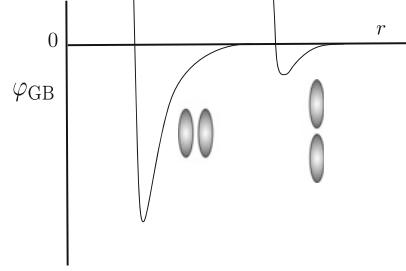


Fig. 1.12 The Gay-Berne potential depends strongly on the relative orientations of the rigid bodies, after a figure in [143]



in providing a precise parameterization [187]: using a general model for water (one not optimized for their specific experiment) the melting point of ice was underestimated by approximately 20 °K.

1.2 The N-Body Problem

In this book, we shall frequently use a compact, vectorial notation, where $\mathbf{q}$ and $\dot{\mathbf{q}}$ represent vectors of the positions and velocities, and $\mathbf{M}$ is a diagonal mass matrix, so the equations of motion (1.2) become

$$\mathbf{M} \frac{d^2}{dt^2} \mathbf{q} = \mathbf{F}(\mathbf{q}) = -\nabla U(\mathbf{q}). \quad (1.3)$$

If the N particles are confined to the real line, then $\mathbf{q} = (q_1, q_2, \dots, q_N)$ and the components of the force $\mathbf{F}$ are $F_i = -\partial U / \partial q_i$. For motion in $\mathbb{R}^3$, we write $\mathbf{q}_i = (q_{i,x}, q_{i,y}, q_{i,z})$ for the position vector of the i th particle, and the force acting on this particle is $\mathbf{F}_i = -\nabla_{\mathbf{q}_i} U = -\partial U / \partial \mathbf{q}_i$, that is, the vector with components $F_{i,x} = -\partial U / \partial q_{i,x}$, $F_{i,y} = -\partial U / \partial q_{i,y}$, $F_{i,z} = -\partial U / \partial q_{i,z}$. We may also write x_k, y_k, z_k for the x, y and z components of the position of particle k . We let N_c represent the total number of position coordinates of our system, thus for a system of N atoms in three space dimensions there are $N_c = 3N$ coordinates. In the typical case of 3-dimensional space, we view $\mathbf{q}$ as a $N_c = 3N$ -dimensional vector and write $\mathbf{M} = \text{diag}(m_1, m_1, m_1, m_2, m_2, m_2, \dots, m_N, m_N, m_N)$.

The number of local directions in which the configurational (position) state can be varied is called the number of *degrees of freedom* N_d . For the N -body system in $\mathbb{R}^3$ without additional constraints there are $N_d = N_c = 3N$ degrees of freedom. If r independent constraints are present the number of degrees of freedom is $N_d = N_c - r$.

The total energy of the N -body system is a function of positions and velocities,

$$E(\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_N, \dot{\mathbf{q}}_1, \dot{\mathbf{q}}_2, \dots, \dot{\mathbf{q}}_N) = \sum_{j=1}^N \frac{m_j \|\dot{\mathbf{q}}_j\|^2}{2} + U(\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_N). \quad (1.4)$$

This is not a big step from the Newtonian form of the equations of motion $\dot{\mathbf{q}} = \mathbf{v}; \mathbf{M}\dot{\mathbf{v}} = \mathbf{F}$. More generally, for molecular models, the Hamiltonian and Lagrangian formulations are interchangeable, but the use of the Hamiltonian form is preferred for allowing simplified description of the geometric character of the solutions of the system as we discuss in Chaps. 2–4.

The set of all positions and momenta for which the energy is finite is termed the *phase space*. The instantaneous state of a molecular system involving many, say N , particles moving in $\mathbb{R}^3$ is described by coordinates and positions, i.e., by a point in $\mathbb{R}^{6N}$. The “points” defining atomic positions (generally taken to be located in $\mathbb{R}^3$) should be clearly distinguished from the “points,” in $\mathbb{R}^{6N}$, of the phase space, which represent the collection of all position and momenta vectors of the atoms. One may speak of a “phase space point” or simply “phase point” to describe the latter. We are ultimately going to be interested in how sets of phase points are carried forward by solution of the differential equations. It is crucial to observe that when we discuss such sets the individual points therein are understood to be completely independent of one another, and their motion is governed only by the laws of mechanics as described by the equations of motion.

1.5 Existence and Uniqueness of Solutions

One of the most important properties of a typical classical molecular Hamiltonian system is the existence and uniqueness of solutions started from a generic initial condition. The traditional study of existence and uniqueness for systems of nonlinear ordinary differential equations gives local results only, but for Hamiltonian systems considered in molecular dynamics the solutions are typically globally defined. The uniqueness of solutions is easily verified in the usual way (as for the local result for uniqueness of solutions). The key point is that, with the energy constraint, solutions typically remain bounded for all time.

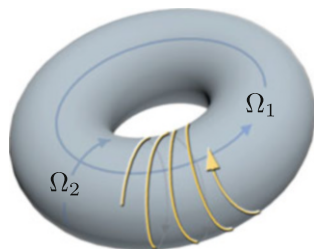
Assume that U is a potential energy function which is bounded below, $U \geq U_{\min}$. For given $E_0 \geq U_{\min}$ define $\Sigma_{E_0} = \{(\mathbf{q}, \mathbf{p}) | H(\mathbf{q}, \mathbf{p}) = E_0\}$. Then, for $(\mathbf{q}, \mathbf{p}) \in \Sigma_{E_0}$ we have

$$\frac{\mathbf{p}^T \mathbf{M}^{-1} \mathbf{p}}{2} + U(\mathbf{q}) = E_0 \Rightarrow \frac{\mathbf{p}^T \mathbf{M}^{-1} \mathbf{p}}{2} = E_0 - U(\mathbf{q}) \leq E_0 - U_{\min}.$$

$\mathbf{M}^{-1}$ is a positive definite matrix (assumed here to be constant), so we can infer that the momenta are bounded at fixed total energy. We would like to say something similar for positions. We have, at energy E_0 ,

$$U_{\min} \leq U(\mathbf{q}) \leq E_0.$$

Fig. 1.14 Integrable dynamics may be viewed as periodic or quasi-periodic motion on the surface of a torus (illustrated for a system with two degrees of freedom)



1.5.3 Equilibrium Points, and the Linearization Theorem

Molecular systems can be written in the form

$$\frac{dz}{dt} = f(z). \quad (1.6)$$

An *equilibrium point* of such a system is a solution of $f(z) = 0$.⁴ An equilibrium point z^* corresponds to an equilibrium solution, since if we define a constant function $z(t) \equiv z^*$ then we have $\dot{z}(t) \equiv f(z^*) = 0$. In some cases, it is possible to discuss the motion of solutions close to equilibrium points by means of *linearization*: the technique of replacing a nonlinear system by a linear one which approximates it locally.

We assume that f is continuously differentiable in the vicinity of the equilibrium point z^* and make use of the fact that $f(z) \approx f(z^*) + f'(z^*)(z - z^*)$ for $\|z - z^*\|$ sufficiently small. Since $f(z^*) = 0$ we have, defining $\delta z := z - z^*$,

$$\frac{d\delta z}{dt} \approx A\delta z, \quad A = f'(z^*).$$

The symbol $\approx$ is not very precise. In general, the approximation of solutions is only local in time meaning that solutions of the nonlinear system and the linear approximation may diverge. We refer to the system

$$\frac{d\delta z}{dt} = f'(z^*)\delta z$$

as the *linearization* of the system (1.6) around z^* .

In case the equilibrium point is *hyperbolic*, meaning that the real parts of the eigenvalues of $A = f'(z^*)$ are nonzero, then one can infer that the solutions of the

⁴For readers conversant with concepts of statistical mechanics and in preparation for later discussions, we emphasize that the concept of *mechanical equilibrium* given in relation to the system of ordinary differential equation (1.6) is entirely different from the notion of *statistical* or *thermal equilibrium* of Chap. 6.

nonlinear and linear systems are in fact *topologically conjugate*: if $\mathbf{z}$ is the solution of the nonlinear system and $\delta\mathbf{z}$ is the solution of the linear system, then there is a smooth, invertible map Φ of $\mathbb{R}^m$ defined in a neighborhood of the origin such that

$$\mathbf{z}(t) = \mathbf{z}^* + \Phi(\delta\mathbf{z}(t)).$$

This is referred to as the Hartman-Grobman theorem (for more discussion, see [177], where this result is referred to as the “Linearization Theorem”; a proof may be found in [362]).

Let $\mathbf{z}^*$ be an equilibrium point. We say that $\mathbf{z}^*$ is stable (“in the sense of Lyapunov”) if, for all ε , there exists δ such that, for all $\mathbf{z}_0$ such that $\|\mathbf{z}_0 - \mathbf{z}^*\| < \delta$,

$$\sup_{t \geq 0} \|\mathcal{F}_t(\mathbf{z}_0) - \mathbf{z}^*\| < \varepsilon.$$

That is, the solution started near to $\mathbf{z}^*$ stays near $\mathbf{z}^*$ for all time. The Hartman-Grobman theorem clearly implies that the stability of a given hyperbolic equilibrium point $\mathbf{z}^*$ of a nonlinear system can be inferred from the stability of the origin for the linearization of the system around $\mathbf{z}^*$.

In the case of Hamiltonian systems, the eigenvalues are, very often, on the imaginary axis, and the Hartman-Grobman theorem is not of much use. However, there is another result which is relevant in this case. Observe that an equilibrium point $\mathbf{z}^* = (\mathbf{q}^*, \mathbf{p}^*)$ of a Hamiltonian system in “kinetic plus potential” form

$$H(\mathbf{q}, \mathbf{p}) = \mathbf{p}^T \mathbf{M}^{-1} \mathbf{p} / 2 + U(\mathbf{q})$$

will always have $\mathbf{p}^* = 0$ and $\nabla U(\mathbf{q}^*) = 0$. We say that $\mathbf{q}^*$ is a *strong local minimum* of the potential if there exists $\varepsilon > 0$ such that

$$0 < \|\mathbf{q} - \mathbf{q}^*\| < \varepsilon \Rightarrow U(\mathbf{q}) > U(\mathbf{q}^*).$$

Then we have

Theorem 1.1 *If $\mathbf{q}^*$ is a strong local minimum of smooth potential U then $\mathbf{z}^* = (\mathbf{q}^*, \mathbf{0})$ is stable.*

The proof of this theorem relies on showing that if trajectories are started from a point sufficiently close to $\mathbf{z}^*$ they cannot wander away to infinity. Although the result holds in greater generality, it is easy to show under assumptions of local smoothness of U (which we are normally happy to make in molecular dynamics). For more discussion see the text [216]. If the potential is C^2 , the linearized version of this system has “kinetic plus potential” form with Hamiltonian

$$\bar{H} = \frac{\delta \mathbf{p}^T \mathbf{M}^{-1} \delta \mathbf{p}}{2} + \frac{\delta \mathbf{q}^T U''(\mathbf{q}^*) \delta \mathbf{q}}{2}.$$